Investigation of Temperature-Dependent Carrier Scattering Time in Metals via Hall Effect and Resistivity Measurements

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Abstract

In this study, within the framework of the classical Drude transport model, the relationship between the carrier scattering time and temperature was investigated in four metal materials (copper, aluminum, iron, and nickel). The Hall coefficient and longitudinal resistivity were measured within the temperature range of 253K to 323K. For ferromagnetic metals (iron and nickel), the method of magnetic field scanning was adopted to distinguish the ordinary Hall component from the anomalous Hall contribution.

The research results show that for all materials, the resistivity increases monotonically with the increase in temperature, thereby reducing the extracted scattering time. After normalization, all samples exhibit a continuous decrease in scattering time as the temperature rises, indicating that electron-phonon scattering is the dominant temperature-dependent mechanism within this temperature range.

Although copper has a relatively small Hall coefficient, leading to larger uncertainty, the temperature-dependent trend remains experimentally resolvable. Within the temperature range studied, the classical Drude model can provide a physically consistent description of the observed transport behavior.

Key words: Hall effect; carrier scattering time; electron–phonon scattering; electrical resistivity; Drude model; temperature dependence

1. Introduction:

To understand the electrical conduction properties of metals, a clear description of charge carriers and their microscopic dynamics is critical, as is the significant empirical evidence for delocalized electron behavior in interpreting macroscopic transport phenomena. In a metallic conductor, electrical current is mainly carried by delocalized electrons in their drift motion. These microscopic characteristics of charge carriers determine macroscopic transport properties such as electrical resistivity and conductivity. Thus, investigating the behavior of charge carriers suggests that fundamental insight into microscopic transport mechanisms is achievable [1].

As electrons travel through the lattice structure of a metallic conductor, they will inevitably encounter collisions with impurities, crystal defects, grain boundaries, and lattice thermal vibrations (phonons) [1]. These critical scattering processes significantly alter the momentum of delocalized electrons, which are collectively referred to as scattering mechanisms [2]. Moreover, these processes significantly disrupt the directed motion of electrons, which demonstrates that even highly conductive metals exhibit finite resistivity [1,2]. Furthermore, scattering mechanisms indicate that the interaction with the phonons limits the ability of electrons to conduct electrical current.

The scattering time (τ) is defined as the average time interval between successive scattering events, which is a critical parameter that substantially connects microscopic mechanisms of charge carriers to macroscopic transport behavior such as electrical conductivity [2]. Therefore, the classical Drude model suggests that electrical conductivity is directly proportional to the scattering time τ, which is indicated by how long carriers maintain directed motion before scattering [1].

Among various scattering mechanisms, electron-phonon scattering is typically dominant under finite temperature conditions [2], and this critical mechanism plays an important role in temperature-dependent transport of metals. Phonons are defined as the quantized lattice vibrations. Moreover, as temperature increases, thermal vibration of the lattice structure will intensify [2], which demonstrates that electrons will experience more collisions with phonons,

leading to stronger scattering and reduced average time intervals between successive scattering events. Within a moderate temperature range, the temperature dependence of the scattering time is observable as the intensification of the thermal vibration.

In 1879, American physicist Edwin Hall was conducting research on the conductivity of metals when he designed a targeted experiment, where he placed a metal sheet with a constant longitudinal current flowing through it in a magnetic field that was perpendicular to the direction of the current. and he observed an unexpected voltage on the two sides of a metal sheet perpendicular to both current and magnetic field. The voltage between the two edges of the metal sheet was proportionate with the current and the magnetic field. This experiment was later named as the "Hall Effect", which has since become a standard tool for characterizing carrier properties in conductive materials.

Although the electron was formally discovered by J. J. Thompson 18 years later, the observation of the Hall Effect provides early experimental evidence that electrical current in the metal conductor is governed by negatively charged carriers, which played an essential role in confirming the electronic charge transport. This discovery clarified that the charge carriers in the conductor would deflect in the magnetic field, which was the first time that the movement characteristics of charge carriers in a magnetic field were observed through experimental means, and became one of the first tools that can link the macroscopic electrical measurements with microscopic carrier properties.

This study aims to investigate the temperature dependence of scattering time in several metallic materials through the Hall Effect. The scattering time τ can be extracted from the Hall Coefficient and the resistivity of different materials [4, 5, 6] since mobility is proportional to scattering time. This method has been validated in previous studies of combining the Hall Effect and resistivity to extract reliable mobility and τ values [3,9]. To improve the reliability of the measurement, a magnetic field sweeping method is applied, so that the Hall Coefficient of magnetic metal can be determined through linear fitting. Furthermore, both non-magnetic metals (Cu, Al) and magnetic metals (Fe, Ni) are examined for the study. In a moderate temperature range from 253K to 323K, the trend of temperature dependence of scattering time could be observed. It is hypothesized that scattering time will decrease with increasing temperature in the

aimed temperature range, and the analysis based on the Hall Effect suggests that this trend could be effectively captured across different metallic materials [4, 5, 6].

2. Literature Review:

Notwithstanding the important theoretical foundations already established, understanding electrical conduction in metals could require not only basic transport models but also important theoretical frameworks that indicate how scattering mechanisms and measurement techniques capture carrier dynamics at different temperatures. Moreover, the classical models, such as the Drude theory, provide important foundations by demonstrating that electrons appear as free particles whose motion is interrupted by collisions and connecting scattering time to electrical conductivity [1,2]. Thus, real metals present more complex behavior due to multiple scattering sources and temperature-dependent interactions, which require more rigorous theoretical and experimental studies beyond simple assumptions [1,2].

A recent study by Palenskis and Jonkus indicates that electron transport in metallic conductors reveals that deviations from ideal free-electron behavior appear to emerge from critical microscopic scattering processes [1]. Moreover, their study's theoretical findings indicate that even in simple metals, the interaction between random carrier motion and lattice properties could substantially influence the observed transport results. Thus, experiments require measuring not only conductivity but specific mechanisms controlling momentum relaxation. This understanding helps separate phonon-mediated scattering from impurity effects when analyzing temperature dependence [1].

The electron-phonon interaction is a dominant scattering mechanism in metals at finite temperatures, especially above cryogenic ranges. Amarel, Belitz, and Kirkpatrick suggest that rigorous theoretical findings for electrical conductivity caused by electron-phonon scattering demonstrate that temperature strongly influences the rate of momentum-relaxing collisions [2]. The increasing thermal energy raises the phonon population and reduces scattering time τ. Therefore, their discovery supports the common experimental observation that resistivity in simple metals increases with temperature, encouraging the extraction of τ from measured resistivity and Hall coefficient data [2].

Although electron-phonon scattering represents the key focus, other processes, such as structural defects, impurity scattering, and electron-electron interactions, suggest relevant complexity under certain conditions. The studies on semimetals and oxide systems demonstrate that defect scattering or band structure features could substantially influence transport behavior, especially when phonon contributions appear relatively weak [6,7,8]. Thus, multiple scattering channels can be distinguished using combined transport measurements. Although these systems are different from ordinary metals, their methods show how multiple scattering channels can be distinguished using combined transport measurements. As a result, this comparative evidence supports the use of comprehensive measurement plans in the experiment design, including temperature control and multi-parameter analysis [6,7,8].

The Hall effect represents one of the most critical tools for studying carrier dynamics, demonstrating that it connects macroscopic electrical measurements with microscopic properties such as carrier concentration and mobility. Additionally, in the ideal free-electron picture, the Hall coefficient is often treated as a constant determined by carrier density. Early work by Kimura and Honda shows that the temperature dependence of the Hall coefficient cannot be explained by a simple model [3]. Thus, changes in scattering rate and electronic structure lead to observable changes in Hall response, proving that measurements provide important information about scattering behavior [3].

Theoretical studies by Yakovenko and Zheleznyak discuss the temperature dependence of the Hall coefficient in low-dimensional metals, emphasizing how relaxation and anisotropic band structure depart from classical predictions [4]. Moreover, the key findings from their research on quasi-one-dimensional systems suggest that the Hall response is sensitive to the details of carrier dynamics. Thus, temperature-dependent scattering will affect how we interpret Hall measurements. These works effectively support using Hall coefficient changes to represent scattering time, especially with resistivity data [4].

Another important part of the Hall effect is the additional complexity that is presented by magnetic metal systems; the important theoretical implications of anomalous Hall effects indicate that such systems require careful analysis. Furthermore, the significant findings of Husmann et al. on anomalous Hall conductivity in ferromagnetic metals suggest that spin-dependent scattering and magnetic order introduce important temperature-sensitive terms to the

Hall response [5]. Thus, anomalous contributions will affect results in ferromagnetic materials like Fe and Ni. A Hall measurement requires careful interpretation in magnetic metals to ensure observed changes reflect scattering time rather than external magnetic effects [5].

Given that methodological considerations can indicate that extracting reliable transport parameters demands systematic approaches, the significant methodological findings of Kajikawa's research on conductivity and Hall effect in n-type CdS demonstrate that simultaneous measurements provide critical advantages in separating multiple transport contributions [9]. Additionally, the method of combining Hall and resistivity measurements suggests that this approach offers relevant theoretical guidance for the present study. Therefore, resistivity and Hall voltage measurements show carrier mobility and scattering time with higher confidence than single-parameter measurements alone might yield less reliable results [9].

These theoretical and experimental studies indicate that the significant empirical evidence collectively establishes a reliable framework for analyzing temperature-dependent transport in metals, demonstrating that

- Carrier scattering mechanisms depend on temperature, and the electron-phonon interaction suggests a key role at moderate temperatures [2].
- The Hall coefficient varies with temperature due to changes in scattering and band structure [3,4].
- Hall Effect and resistivity measurements are effective in obtaining useful scattering parameters [9].
- Magnetic metals show extra complexity requiring distinction between ordinary and anomalous Hall contributions [5].

These findings could indicate that a systematic framework is essential. This study suggests that investigating the temperature dependence of scattering time in non-magnetic and magnetic metals through combined Hall effect and resistivity measurements under controlled conditions can demonstrate that such an approach yields reliable and significant results.

3. Methodology:

The purpose of this study is to investigate the temperature dependence of scattering time τ in the selected metallic materials by combining the Hall Effect and the resistivity of the material. To achieve this, a carefully controlled experiment is conducted to record the Hall Coefficient, and the longitudinal resistivity is measured over a defined temperature range between 253K and 323 K. The Hall Coefficient of Cu and Al is measured at a representative temperature point, 298K, and assumed to be weakly temperature-dependent within the temperature range. The Hall Coefficient of Fe and Ni is measured through the magnetic field sweeping method to extract the ordinary Hall Coefficient due to the existence of the anomalous Hall effect at 253K and 298K and shows that there is no significant deviation of the Hall Coefficient compared to room-temperature measurements.

3.1 Sample Geometry and Materials

For the metallic materials we investigate in this study: Copper (Cu), Aluminum (Al), Iron (Fe), and Nickel (Ni). Each material is prepared into a cross-shaped geometry (Fig.1), with an 8mm main channel width, 25 mm main channel length, 0.1mm thickness, and 4mm Hall arm width, 6mm Hall arm extension to support both longitudinal current injection and transverse Hall voltage measurement. This configuration allows clear separation between the current channel and the Hall voltage probe.

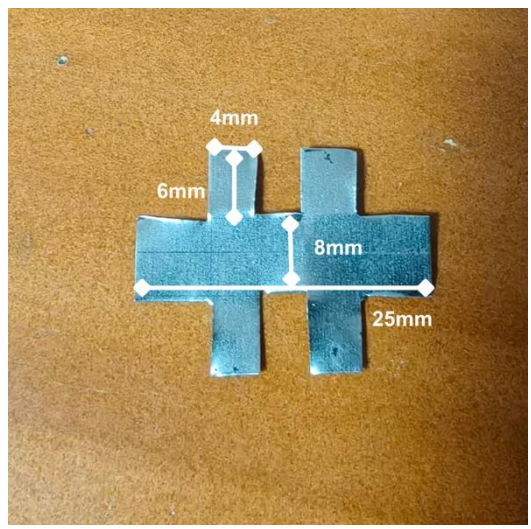


Figure 1. Geometry and dimensions of the cross-shaped metal sample. Thickness = 0.10 mm.

The geometric parameters of each sample are measured using a digital caliper with a resolution of 0.01mm, while the thickness of the samples is provided and labeled by the merchant. The thickness t of the samples plays a critical role in calculating the Hall voltage:

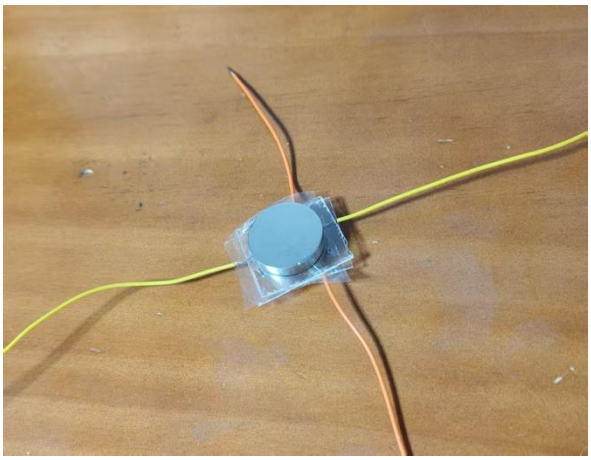


Figure 2.Sample configuration in the magnetic field with four-terminal contacts

$$R_H = \frac{V_H t}{IB}$$

Where V_H is the Hall Voltage, I is the applied current, and B is the magnetic field strength. All samples are measured three times and averaged to reduce the random measurement error. The geometric uncertainty was estimated based on the instrument resolution and repeated measurements.

The sample was placed between two magnets. At the top and bottom of the sample, there were each an acrylic plate to prevent direct contact between the sample and the magnets. Among the four required endpoints, a wire was connected to each for insertion into the Hall effect measurement setup, and the wires were fixed on the sample with transparent tape to ensure stable current conduction (four-terminal contact) (Fig.2).

3.2 Electrical Measurement setup

3.2.1 Hall Voltage setup

To measure the Hall Voltage, a constant current source is used to provide a steady longitudinal current supply (Fig.3,4). The current direction is fixed along the horizontal axis of the main tunnel of the sample, where the current magnitude is monitored by an ammeter. The current variation during each measurement is controllable and can be adjusted by the slide rheostat.

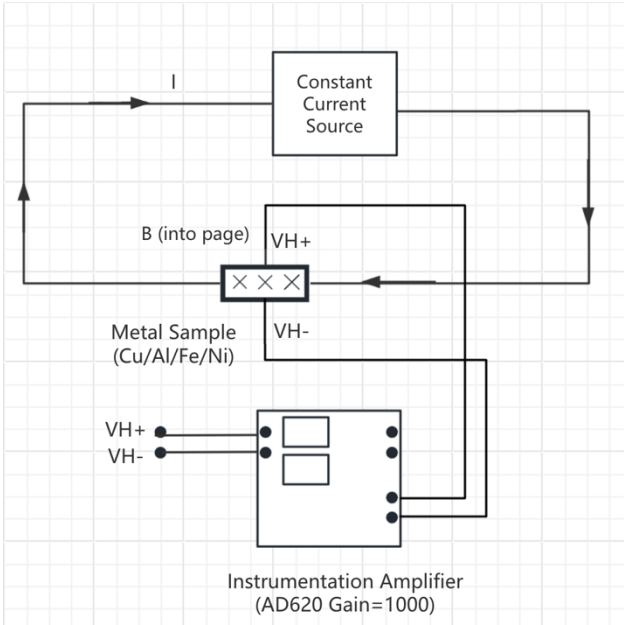


Figure 3. Circuit diagram of the Hall effect measurement setup

A magnetic field is applied perpendicular to the sample and the current direction, where the magnitude of the magnetic field is measured by a Gauss meter. The field contributions from the upper and lower magnets are measured independently, and the total magnetic field is calculated by linear superposition. This approach is justified under the assumption that the magnetic field obeys the principle of superposition in free space.

Under these conditions, the charge carriers experience a Lorentz force, which deviates the direction of the motion of the electrons, and finally leads to the development of a transverse Hall Voltage across the vertical Hall arm of the sample due to the accumulation of electrons on the edge of the sample.

The Hall Voltage is measured using an instrumentation amplifier (AD620) with a gain of 1000 to amplify the microvolt-level signal. The amplifier output voltage is then recorded using a multimeter (Fig.3,4).

To improve measurement reliability, a magnetic field sweep is applied to Fe and Ni since they are magnetic metals, which exhibit the anomalous Hall Effect. In addition, the Hall Voltage is measured for both positive and negative current directions to eliminate thermal noise and other factors:

$$V_H = \frac{V(+I) - V(-I)}{2}$$

A Linear fitting of V_H versus B is applied to determine the Hall Coefficient from the slope. This method reduces systematic offsets and improves the extraction accuracy of the ordinary Hall component.

Figure 4. Experimental setup for Hall voltage measurement (practical implementation)

For ferromagnetic metals (such as iron and nickel), at low magnetic field conditions, the Hall voltage contains both the ordinary Hall effect contribution and the anomalous Hall effect contribution. In the low magnetic field region, due to the unsaturated magnetization effect, the anomalous Hall effect leads to significant nonlinear characteristics in the $V_H - B$ relationship.

To extract the conventional Hall coefficient, only the high magnetic field region with magnetization close to saturation was selected for linear fitting. Specifically, data points ranging from 500 mT to 700 mT were used for fitting, while the low magnetic field region (300 mT to 500 mT) was excluded from the analysis as it exhibited obvious nonlinear characteristics.

This procedure ensures that the extracted slope mainly reflects the ordinary Hall component.

3.2.2 Resistivity setup

The electrical resistivity of each metallic sample is determined from the longitudinal voltage drop measured along the current direction. A constant current I is applied through the two opposite arms of the cross-shaped sample, and the longitudinal voltage V is measured between the voltage probe aligned with the current direction. Since the signal of the voltage is microvolts, the instrumentation amplifier will be applied in the measurement of resistivity with the same scale as the measurement of Hall Voltage.

The electrical resistance is calculated using:

$$R = \frac{V}{I}$$

The resistivity ρ is then obtained from the geometrical relation:

$$\rho = R\frac{L}{A}$$

Where L is the distance between the two longitudinal voltage probes, and A is the cross-sectional area of the sample $A = \omega t$. ω is the width of the main tunnel of the sample, and t is the thickness of the sample.

Substituting the expressions gives:

$$\rho = \frac{V}{I}\frac{\omega t}{L}$$

All these resistivities are measured in the targeted temperature point (253K, 273K, 298K, 313K,323K) to ensure consistency when extracting the scattering time from R_H and ρ values.

3.3 Temperature control system

The temperature dependence of scattering time is measured between 253K and 323K. There are two temperature control methods applied in this experiment

3.3.1 Mid-Temperature Region (Water Bath)

For the temperature region that is higher than room temperature (313K, 323K), a water bath system has been used. The sample is placed in a sealable bag to avoid contact and immersed in the hot water. The temperature is monitored by a digital thermometer with 0.1 resolution, which is placed on the sample, to ensure the sample reaches the target temperature. The measurements are conducted only after temperature stabilization for 2 minutes (Fig.5).

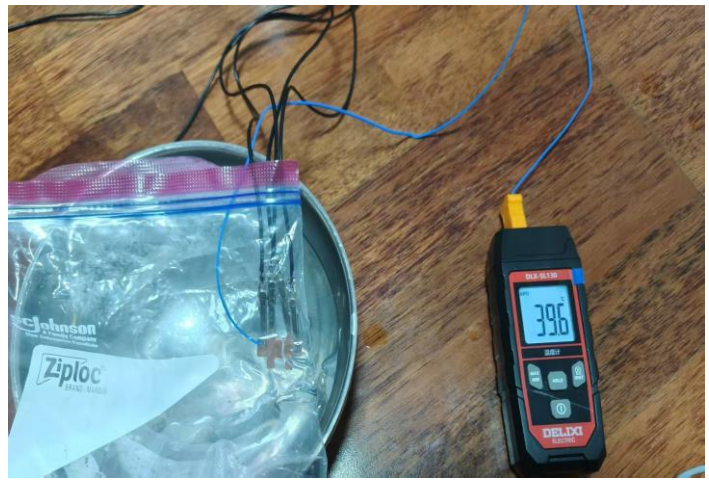


Figure 5.Elevated-temperature measurement setup using a water bath

3.3.2 Low-Temperature Region (Cooling Pack Method)

For the temperature region that is lower than the room temperature (253K, 273K), a sealed cooling pack system is applied. The sample is placed inside a cooling bag, with a towel surrounding the sample (Fig.6).

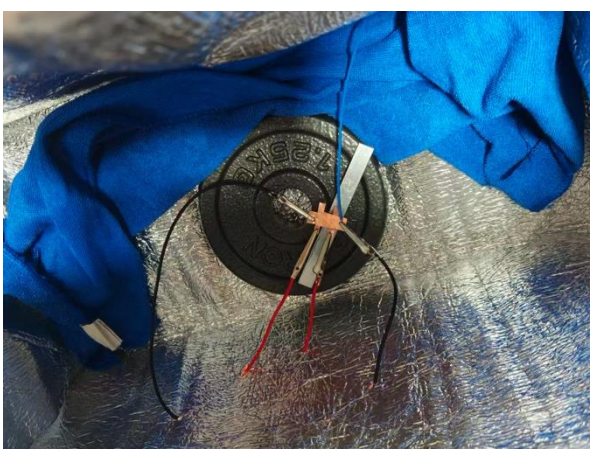


Figure 6. Low-temperature measurement setup using an insulated cooling pack

To sustain a low temperature for a longer time, a cold plate (1.25KG dumbbell sheet) is placed under the sample, with an aluminum bar placed between the sample and the cold plate, where the probe of the thermometer is placed adjacent to the sample to capture real-time temperature. To achieve this temperature, a fast refrigerant is used to reach the target temperature. The cold plate and the aluminum bar could sustain the target temperature for approximately 30 seconds, during which the measurement can be completed (Fig.6).

Although this method is not a cryogenic system, it provides a stable and controllable low-temperature environment sufficient to observe the temperature dependence of resistivity within the targeted temperature range.

3.4 Assumption and Experimental Consideration

Several assumptions are adopted in this study:

- For non-magnetic metals (Cu, Al), the Hall Coefficient is assumed to be weakly temperature-dependent within the temperature range.
- For magnetic metals (Fe and Ni), magnetic field sweeping measurements are applied at low temperature (253K) to determine the Hall coefficient through linear fitting. The extracted Hall coefficients show that there is no significant deviation compared to room-temperature measurements.
- Contact resistance is assumed to be negligible compared to the sample resistance.
- Magnetic field uniformity is assumed within the effective sample area, and the magnetic field obeys the linear superposition in free space.
- The classical Drude model framework is used to relate mobility and scattering time.
- The effective mass is assumed to be the static electron mass.

Under the classical Drude model framework, electrical conductivity σ is given by:

$$\sigma = \frac{ne^2\tau}{m}$$

Where n refers to carrier concentration, e refers to elementary charge, τ refers to scattering time, m refers to electron mass.

Since resistivity $\rho = \frac{1}{\sigma}$ and the Hall Coefficient R_H in the free-electron approximation are:

$$R_H = \frac{1}{ne}$$

The scattering time can be obtained by combining these relations:

$$\tau = \frac{m}{e}\frac{R_H}{\rho}$$

Where m, e is constant, and the temperature dependence of R_H in the defined temperature range, so the temperature dependence of scattering time is:

$$\tau(T) = \frac{mR_H}{e}\frac{1}{\rho(T)}$$

3.5 Uncertainty Analysis

All uncertainties are evaluated using high-school level approach. For quantities measured repeatedly, the uncertainty is estimated by the half-range method.

$$\Delta x = \frac{x_{max} - x_{min}}{2}$$

If the instrument resolution is comparable to or larger than the observed fluctuation, the instrument resolution is taken as the uncertainty. Since the Hall voltage is at the microvolt level and close to the resolution limit of the measurement system, potential systematic effects from the voltmeter and the instrumentation amplifier were not included in the uncertainty analysis; they are discussed separately in the Discussion section.

3.5.1 Hall Voltage Uncertainty

At each measurement condition, the Hall Voltage V_H is recorded repeatedly. The representative value was taken as the average:

$$\Delta V_H = \frac{1}{n}\sum_{i=1}^{n} V_{H,i}$$

The uncertainty of the Hall Voltage is estimated using the half-range method:

$$\Delta V_H = \frac{V_{max} - V_{min}}{2}$$

3.5.2 Magnetic Field Uncertainty

The magnetic field at the sample position is calculated by measuring the field produced by the upper and lower magnets separately and applying linear superposition:

$$B = B_1 + B_2$$

The magnetic field uncertainty is estimated as:

$$\Delta B = \Delta B_1 + \Delta B_2$$

With $\Delta B_1 = \Delta B_2 = 0.01mT$ (the resolution of the Gauss meter), the total uncertainty is:

$$\Delta B = 0.02mT$$

Since the magnetic field magnitude is 300mT to 700mT, the relative uncertainty at 300mT is:

$$\frac{\Delta B}{B} \approx 6.67 \times 10^{-5}$$

This contribution is significantly smaller than the relative uncertainty of the Hall voltage and therefore has a negligible influence on the overall uncertainty of the Hall coefficient.

3.5.3 Current Uncertainty

The applied current is provided by a constant current source and monitored during the experiment. The resolution of the ammeter is 0.02A, so:

$$\Delta I = 0.02A$$

3.5.4 Thickness Uncertainty

The thickness of the metal sheets is specified by the supplier as 0.10 mm with a tolerance of ±10%. Therefore, the thickness uncertainty is taken as:

$$\Delta t = 0.10 \times 10\% = 0.01mm$$

Since no independent thickness calibration measurement is performed, the supplier tolerance is adopted as the uncertainty estimate.

3.5.5 Uncertainty of Hall Coefficient

The Hall Coefficient is calculated by:

$$R_H = \frac{V_H t}{IB}$$

Since the Hall coefficient is calculated from a product and a quotient of measured quantities, the relative uncertainty was estimated by summing the relative uncertainties of each variable:

$$\frac{\Delta R_H}{R_H} = \frac{\Delta t}{t} + \frac{\Delta I}{I} + \frac{\Delta V}{V} + \frac{\Delta B}{B}$$

Thus,

$$\Delta R_H = R_H \times \frac{\Delta R_H}{R_H}$$

3.5.6 Uncertainty of Resistivity

The resistivity is calculated by:

$$\rho = \frac{V}{I}\frac{\omega t}{L}$$

So, the uncertainty of resistivity is:

$$\frac{\Delta \rho}{\rho} = \frac{\Delta V}{V} + \frac{\Delta I}{I} + \frac{\Delta L}{L} + \frac{\Delta t}{t} + \frac{\Delta \omega}{\omega}$$

Using the sample linear uncertainty propagation method.

3.5.7 Uncertainty of Scattering Time

The scattering time is calculated from:

$$\tau = \frac{m}{e}\frac{R_H}{\rho}$$

Because m and e are constant, the relative uncertainty of τ is:

$$\frac{\Delta \tau}{\tau} = \frac{\Delta R_H}{R_H} + \frac{\Delta \rho}{\rho}$$

4. Result

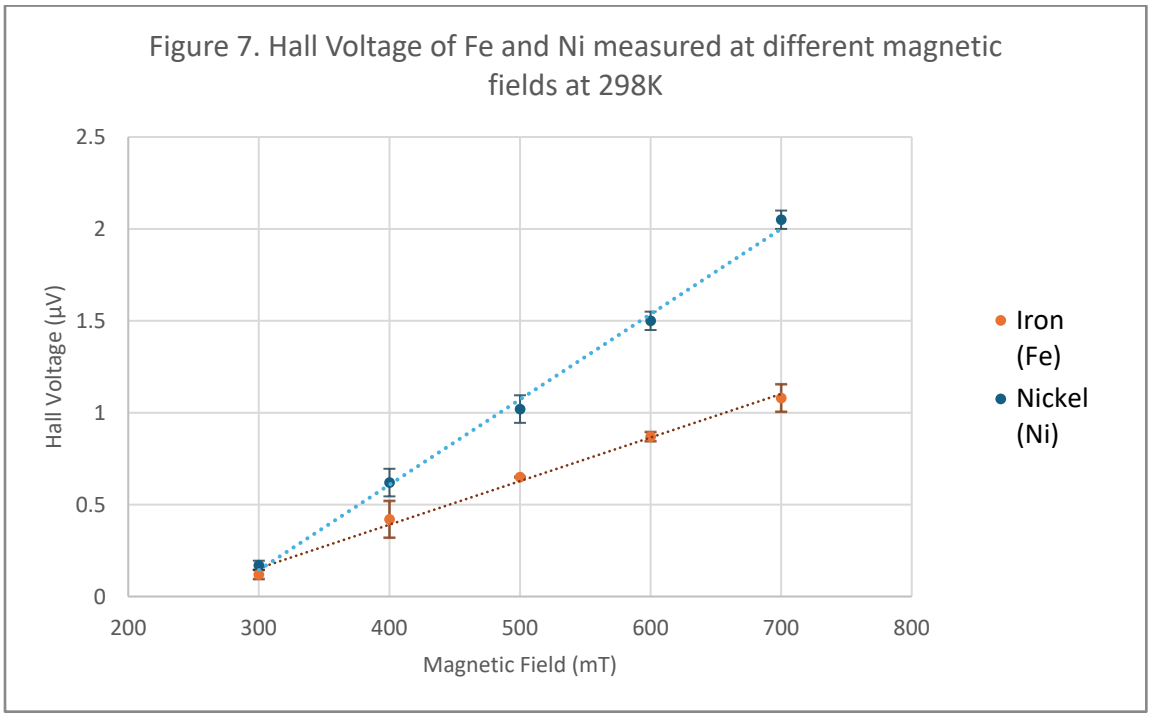
4.1 Hall Voltage and Hall Coefficient

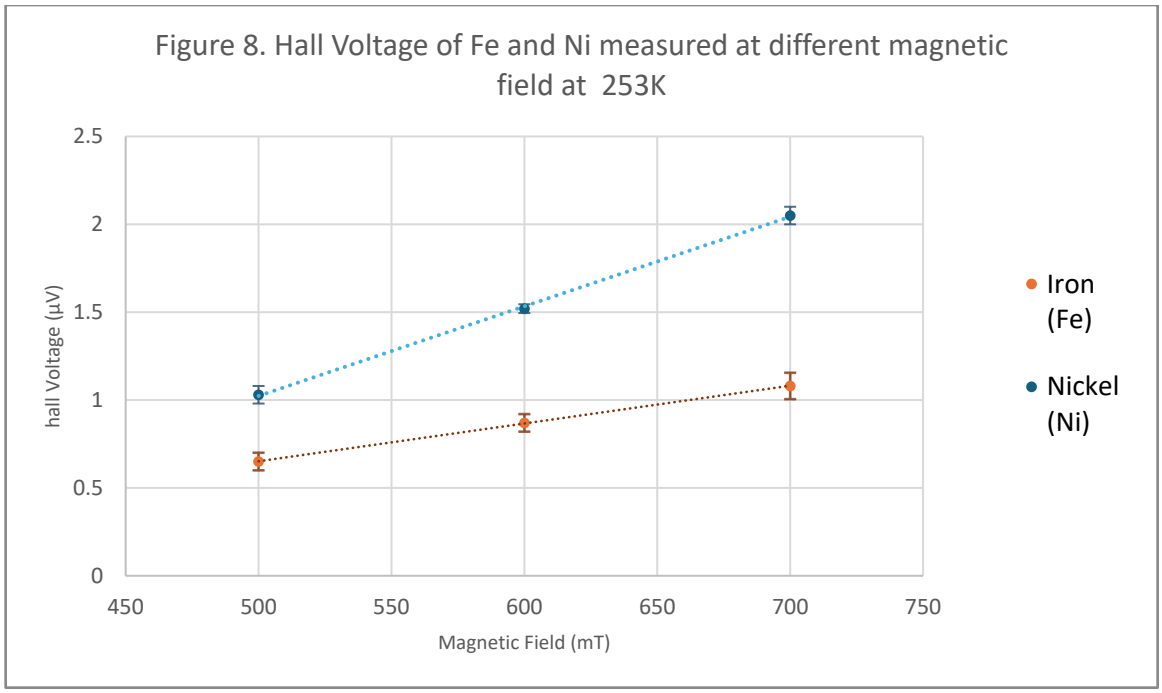
The Hall voltage is measured under a perpendicular magnetic field for all four metallic samples. For non-magnetic metals (Cu and Al), measurements are performed at 298K under a fixed magnetic field.

For magnetic metals (Fe and Ni), the magnetic sweeping method is applied at 253K to 298K to separate the ordinary Hall component from the anomalous Hall contribution. The Hall voltage was measured under magnetic fields ranging from 300 mT to 700 mT.

Figures 7 and 8 show the Hall voltage as a function of magnetic field at 298 K and 253 K, respectively.

In both materials, the $V_H - B$ relationship exhibits approximately linear behavior in the high-field region. Linear fitting was performed using data between 500 mT and 700 mT. The extracted slopes were used to determine the ordinary Hall coefficient.





At 253K, the linear fitting yields:

- Fe: slope= $(2.15 \pm 0.10) \times 10^{-6}$ V/T
- Ni: slope= $(5.15 \pm 0.20) \times 10^{-6}$ V/T

At 298K, the fitted slopes are:

- Fe: slope= $(2.15 \pm 0.05) \times 10^{-6}$ V/T
- Ni: slope= $(5.15 \pm 0.35) \times 10^{-6}$ V/T

The Hall Coefficient is calculated using:

$$R_H = \frac{V_H t}{IB}$$

For Fe and Ni, the Hall Coefficient is extracted by using:

$$R_H = \frac{t}{I}\frac{dV_H}{dB}$$

The extracted values R_H are summarized in Table 1.

Table 1. Extracted Hall coefficients $R_H(m^3 \cdot C^{-1})$ of Cu, Al, Fe, and Ni.

Material	Temperature (K)	$R_H(m^3 \cdot C^{-1})$
Cu	298K	$(1.80 \pm 0.60) \times 10^{-10}$
Al	298K	$(3.75 \pm 0.82) \times 10^{-11}$
Fe	253K	$(4.30 \pm 2.90) \times 10^{-10}$
Fe	298K	$(4.30 \pm 0.70) \times 10^{-10}$
Ni	253K	$(1.00 \pm 0.30) \times 10^{-9}$
Ni	298K	$(1.00 \pm 0.20) \times 10^{-9}$

4.2 Temperature Dependence of Resistivity

The longitudinal resistivity of Cu, Al, Fe, and Ni is measured at five temperature points: 253 K, 273 K, 298 K, 313 K, and 323 K.

The resistivity is calculated using the measured longitudinal voltage, current, and sample geometry.

The measured resistivity values are summarized in Table 2.

Table 2. Temperature dependence of resistivity ρ ($\times 10^{-8}\ \Omega \cdot m$).

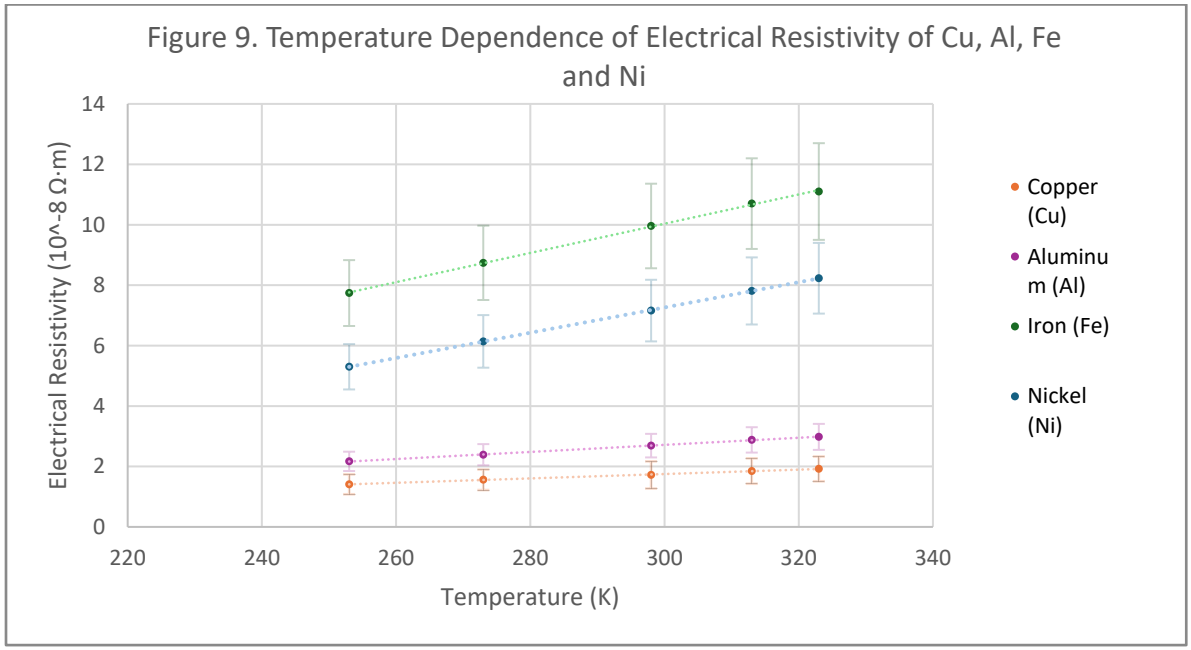
Material	253K	273K	298K	313K	323K
Cu	1.41 ± 0.33	1.56 ± 0.34	1.72 ± 0.45	1.85 ± 0.42	1.92 ± 0.41
Al	2.17 ± 0.32	2.39 ± 0.35	2.69 ± 0.39	2.88 ± 0.42	2.98 ± 0.43
Fe	7.74 ± 1.09	8.74 ± 1.23	9.96 ± 1.40	10.7 ± 1.5	11.1±1.6
Ni	5.30 ± 0.75	6.14 ± 0.87	7.16 ± 1.02	7.81 ± 1.11	8.23 ± 1.17

For all four metallic materials, resistivity increases monotonically with temperature within the investigated range.

For non-magnetic metals (Cu and Al), the resistivity exhibits an approximately linear dependence on temperature between 253 K and 323 K within experimental uncertainty.

For ferromagnetic metals (Fe and Ni), resistivity also increases with temperature. The temperature dependence of resistivity for Fe and Ni is larger than that of Cu and Al, which presents a stronger temperature sensitivity.

Figure 9 shows the temperature dependence of resistivity for Cu, Al, Fe, and Ni



For the relative uncertainty of resistivity, it is calculated as described in Section 3.5. In general, the uncertainty of the resistivity remains smaller than the temperature-induced variation, which indicates that the trend is experimentally resolvable.

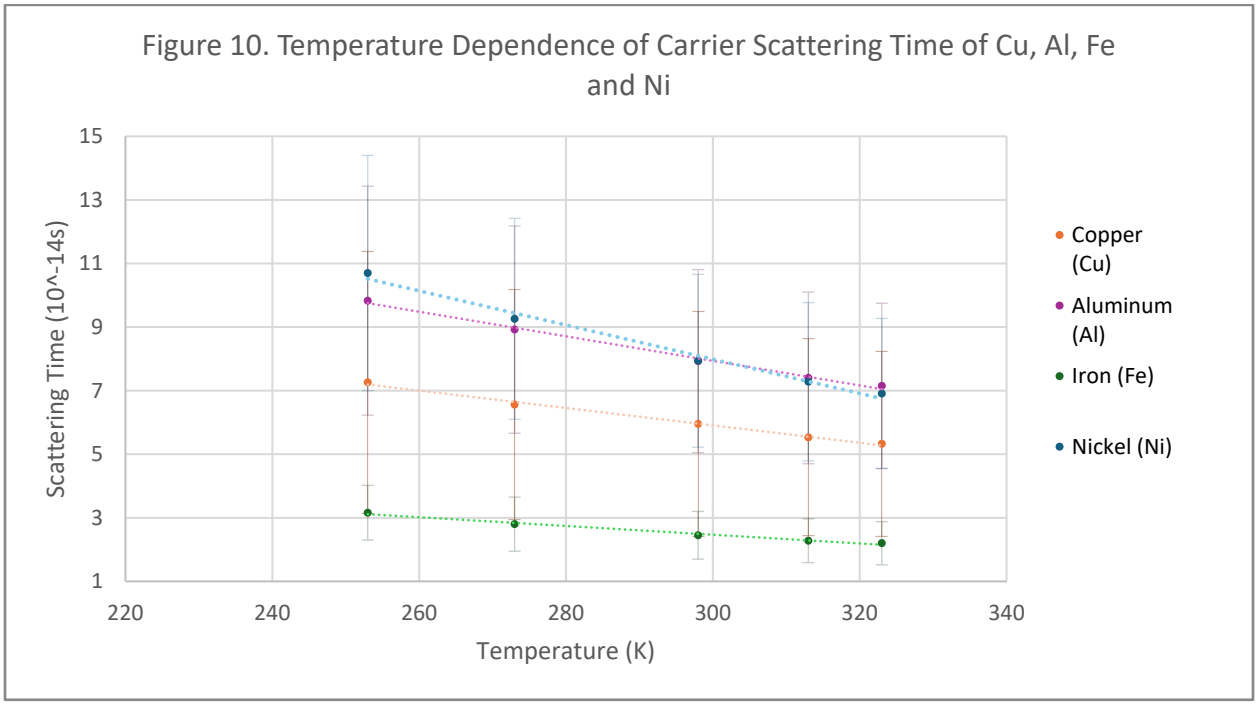
4.3 Temperature Dependence of Scattering Time

The scattering time τ is derived from the combination of the Hall Coefficient and the resistivity of the material as described in Section 3. The extracted values of scattering time are summarized in Table 3.

Table 3. Extracted Scattering Time τ ($\times 10^{-14}$ s) of Cu, Al, Fe, and Ni

Material	253K	273K	298K	313K	323K
Cu	7.26 ± 4.12	6.56 ± 3.62	5.95 ± 3.54	5.53 ± 3.10	5.33 ± 2.91

Al	0.983 ± 0.360	0.892 ± 0.326	0.793 ± 0.288	0.740 ± 0.270	0.715 ± 0.260
Fe	3.16 ± 0.86	2.80 ± 0.85	2.45 ± 0.75	2.28 ± 0.69	2.20 ± 0.68
Ni	10.7 ± 3.7	9.26 ± 3.16	7.94 ± 2.72	7.28 ± 2.49	6.91 ± 2.36



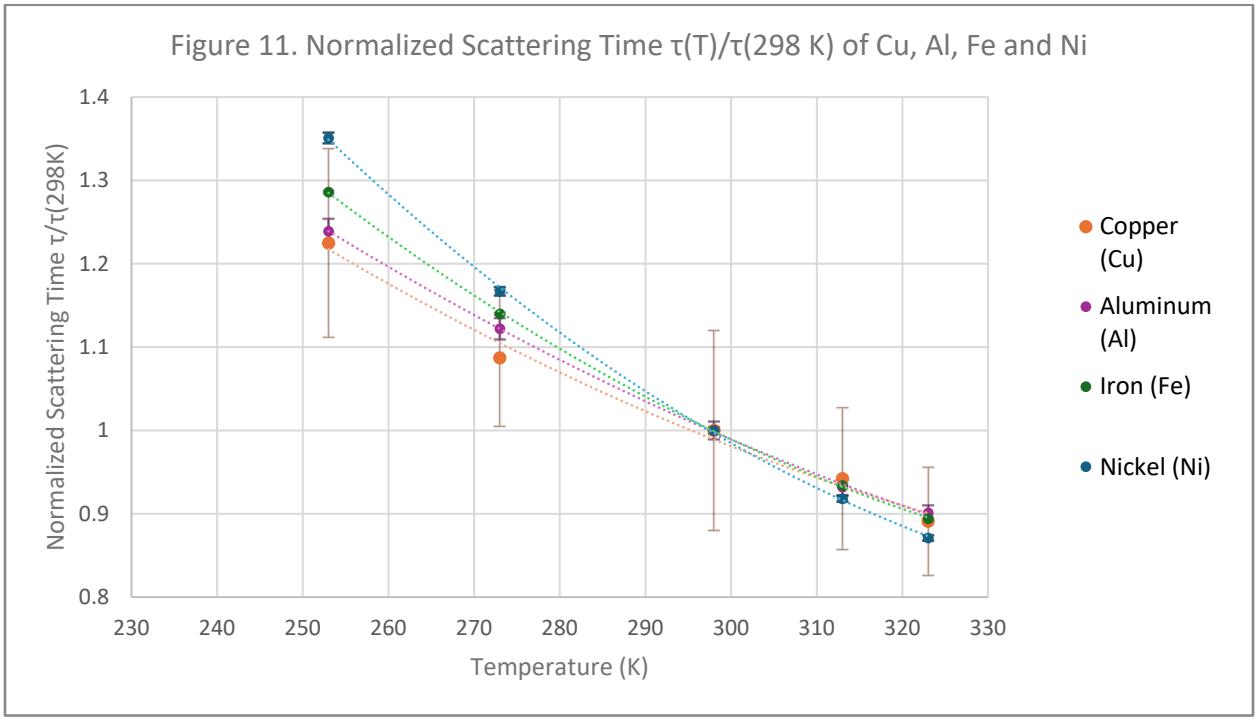
Based on the table and Figure 10, the scattering time of all materials decreases with increasing temperature within the studied range. This behavior is opposite to the temperature dependence of resistivity and reflects the increased frequency of scattering events with the increase in temperature.

For Cu and Al, since the Hall Coefficient is measured at 298K and assumed to have a weak temperature dependence within the studied range, the change of the scattering time is mainly controlled by the temperature dependence of resistivity. As resistivity increases with temperature, the corresponding scattering time decreases.

For Fe and Ni, the scattering time is determined using the Hall Coefficient that is extracted from the high magnetic field sweeping at 253K to 298K. In both materials, the scattering time at 298K is shorter than the scattering time at 253K, which indicates the enhanced scattering at higher

temperature. The relative change corresponds to the increase in resistivity between these two temperatures.

To enhance the comparison among all materials with distinct scattering times, the scattering time is normalized to its value at 298K, which is defined as $\tau_{norm}(T) = \frac{\tau(T)}{\tau(298K)}$. The normalized results are shown in Figure 11.



After the normalization, the curves of all samples showed a consistent downward trend as the temperature increased. This result indicates that the enhancement of the temperature-induced scattering process results in this pattern, rather than the differences in carrier concentrations of different materials.

The relative uncertainty of τ is still smaller than the variation of temperature, which also indicates that the trend of temperature-dependent decrease that we observed is experimentally reliable and resolvable.

5. Discussion

The experimental results show that for all the studied metal materials, within the temperature range of 253K to 323K, the scattering time of the materials becomes shorter as the temperature rises. This phenomenon corresponds to the description of the classical theory of temperature-dependent carrier scattering in metals.

In the Drude model, the scattering time represents the average time interval between two successive momentum relaxation collisions. As the temperature increases, the lattice vibrations become stronger and more intense (the number of phonons increases). Due to the increase in phonons, the probability of electron-phonon scattering also increases accordingly, thus causing the average time interval between collisions to become shorter. Therefore, as the temperature increases, the increased phonon population enhances momentum relaxation collisions with electrons, leading to a reduced scattering time. This behavior is reflected experimentally in the monotonic increase of resistivity and the corresponding decrease in τ.

For non-magnetic metals (copper and aluminum), we measured their Hall coefficients under the condition of 298K and assumed that they have a slight temperature dependence within our study range. In simple metals, the carrier concentration is mainly determined by the valence electrons of the metal and remains basically unchanged within the given temperature range of the experiment. Correspondingly, previous reports indicate that in simple metals, the Hall coefficient changes only weakly with temperature, and its variation is much smaller than the change in resistivity within a similar temperature range [3, 4]. Although the temperature dependence of the Hall coefficient cannot be completely ruled out, its variation is expected to be significantly smaller than that of the resistivity within the current experimental error range. Therefore, the temperature dependence of the scattering time extracted from copper and aluminum mainly comes from the temperature dependence of the resistivity, rather than being determined by the change in the Hall coefficient.

Even though the Hall coefficient does have a slight temperature dependence, its relative contribution to the extracted scattering time is much smaller than the dominant role of the resistivity in the temperature dependence. Therefore, as the temperature rises and the observed scattering time becomes shorter, it becomes less sensitive and unreliable due to the slight change in the Hall coefficient.

Of course, for ferromagnetic metals (iron and nickel), the situation is more complicated due to the appearance of the anomalous Hall effect. In low magnetic fields, the contribution of magnetization to the Hall voltage is significant, resulting in a non-linear behavior. By restricting the linear fitting to the high magnetic field region (500mT - 700mT), the extracted Hall coefficients mainly reflect the ordinary Hall components, and the magnetization phenomenon based on the anomalous Hall effect has approached saturation. The consistency of the extracted Hall coefficients at 253K and 298K for the Hall coefficient indicates that within such temperature ranges, the temperature change does not introduce a large deviation in carrier concentration. Therefore, the observed decrease in scattering time with temperature is due to the enhancement of the scattering mechanism, rather than the change in carrier concentration.

To compare materials with different scattering times, we normalized their scattering times. After normalization, all materials showed a similar trend of scattering time decreasing with temperature. This indicates that although there are certain differences in the absolute scattering times among copper, aluminum, and iron-nickel, the temperature-dependent mechanisms behind them are highly consistent and determined by the same scattering mechanism. This comparison provides such an explanation for the normalization process: within the temperature range of the experiment, electron-phonon scattering is the main temperature-dependent process.

The relatively large uncertainty of copper in the experiment mainly comes from its small Hall Coefficient, which enhances the uncertainty of the scattering time. Nevertheless, the monotonic trend of the decreasing scattering time with increasing temperature remains observable within the experimental resolution. Although the absolute uncertainty of the Hall Coefficient of copper is not negligible, the total variation of scattering time in the defined temperature range exceeds the corresponding uncertainty boundaries. Therefore, the observed temperature-dependent decrease in scattering time can still be experimentally resolved.

From a more detailed theoretical perspective, the Bloch–Grüneisen theory can describe the variation of metal resistivity with temperature [10]. This theory links electron-phonon scattering with lattice vibrations. Within the medium temperature range examined in this study, the resistivity of simple metals theoretically increases approximately linearly with temperature. Since in the Drude framework, the scattering time is inversely proportional to the resistivity, the nearly linear increase in resistivity will also cause the scattering time to be shortened

accordingly. This is consistent with the observed trends for copper, aluminum, iron, and nickel in experiments.

In the classical Drude model, electrons are treated as free carriers with a constant effective mass. However, in actual metals, the band structure and additional scattering mechanisms may cause it to deviate from the assumptions of this simplified model. Nevertheless, within the moderate temperature range studied here, the classical framework can provide a sufficiently accurate explanation for the observed transport behavior that varies with temperature.

The present experiment remains within the framework of the classical Drude model. At sufficiently low temperatures, the deviations from the simple description of the Drude model may become significant, and quantum corrections to conductivity could play a more important role.

6. Conclusion

Based on the classical Drude transport model and the non-quantized, linear response conditions of the Hall effect theory, this study systematically investigates the carrier scattering time and its temperature dependence in four different metallic materials (copper, aluminum, iron, and nickel), conducting experimental research and theoretical analysis. The measurement range is from 253K to 323K.

The experimental results show that in the studied temperature range, the resistivity increases monotonically with the increase in temperature, and thus the scattering time of different materials decreases generally with the increase in temperature, which indicates that electron-phonon scattering gradually becomes the dominant scattering mechanism with the increase in temperature. This experimental result is consistent qualitatively with the Drude carrier transport theory under the condition of electron-phonon scattering dominance. In this description, lattice vibrations increase the probability of momentum relaxation collisions, thereby shortening the time interval between each scattering event.

For non-magnetic metals (copper and aluminum), the weak temperature dependence of the Hall coefficient makes the observed scattering time changes mainly governed by the temperature

dependence of the resistivity. For ferromagnetic metals (iron and nickel), this study applies the method of sweeping the magnetic field to separate the ordinary Hall component, ensuring that even if there is an anomalous Hall effect, the carrier-related parameters can be reliably and efficiently extracted through the resistivity and Hall coefficient.

After normalization, the scattering time of all materials shows a decreasing trend with the increase in temperature, indicating that the carrier scattering mechanism that plays a dominant role in different metals is determined by electron-phonon scattering.

Although there are uncertainties in the experiment (especially for copper), which cannot be ignored, due to the small Hall coefficient, the measured Hall voltage is close to the resolution limit of the instrument, and the corresponding temperature dependence trend can still be distinguished within the uncertainty range. Within the medium temperature range examined in this study, the classical Drude model can provide a physically consistent explanation for the transport behavior.

Future improvement measures may include more precise Hall voltage detection, independent thickness verification, more stable temperature sources, and a wider temperature range to further explore the deviation from the classical model.

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